

Enhancing Safety in Li–S Batteries: Mitigating Thermal Instabilities via Electrolyte and Separator Pair Optimization

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Lithium–sulfur (Li–S) batteries are promising candidates for next-generation energy storage, offering high theoretical energy density (2567 Wh kg^{-1}) and specific capacity (1675 mAh g^{-1}), with environmentally friendly sulfur cathodes. However, their practical deployment faces challenges such as polysulfide shuttling, short cycle life, and high self-discharge—issues exacerbated by thermal instability. Excess heat can degrade performance and trigger thermal runaway, compromising safety. This study investigates the impact of electrolyte and separator selection on thermal behavior, using dimethyl ether, dimethoxy ethane, dimethoxy sulfide, and acetonitrile as electrolytes, paired with boron nitride, titanium dioxide, and polyethylene glycol as separators. Thermal metrics such as peak temperatures (319.025 K – 328.437 K), internal resistance, and rate of heat change are evaluated, along with electrochemical performance (energy stored: 22.027 kJ ; voltage range: 2.38 V to 1.7 V ; electrolyte potential: 93.85 mV to 11.01 mV). The results highlight that strategic electrolyte–separator pairing can effectively mitigate heat generation, advancing the safety and reliability of Li–S batteries in demanding energy applications.

1. Introduction

The future of electric vehicles (EVs) in India is poised for significant expansion, driven by a convergence of government policies, evolving market dynamics and rising consumer environmental awareness. As one of the fastest-growing economies with a rapidly expanding middle class, India represents a promising market for large-scale EV adoption.^[1–3] The Indian government has implemented ambitious policies under schemes such as the Faster Adoption and Manufacturing of Hybrid and Electric Vehicles (FAME) and the National Electric Mobility Mission Plan (NEMMP) 2020, aiming to decarbonize the transportation sector. These initiatives, supported by incentives like subsidies, tax relief, and reduced GST rates, have enhanced the financial

viability of EVs. Concurrently, considerable efforts are underway to develop essential infrastructure such as public charging stations and battery-swapping networks to support their widespread integration. Lithium-ion batteries (LIBs) have been pivotal in enabling the commercial success of EVs due to their high energy density, specific capacity, low self-discharge rate, and long cycle life. Commercialized in 1991, LIBs outperformed earlier chemistries like lead–acid and nickel-based batteries. Lead–acid batteries, though used for auxiliary functions, lacked sufficient specific energy and energy density (up to 40 Wh kg^{-1} and 90 Wh L^{-1}) to power EVs unlike modern LIBs (260 Wh kg^{-1} and 700 Wh L^{-1}).^[4] Ni–Zn batteries offered high discharge rates and thermal stability^[5] but were limited by low energy density and lifespan.^[6] Ni–MH batteries improved upon this with a specific energy of 60 – 120 Wh kg^{-1} ,^[7] yet still trailed LIBs. Metal–air batteries

(e.g., Al–air, Zn–air) promise 3–30 times higher energy densities than LIBs, though their practical use in EVs is still constrained by recharging and stability challenges. Lithium–sulfur (Li–S) batteries have emerged as a highly promising next-generation energy storage technology, primarily due to their theoretical specific energy ($\approx 2600 \text{ Wh kg}^{-1}$), cost-effectiveness, and environmental sustainability. Despite their potential, practical implementation is currently constrained by challenges such as polysulfide shuttling, poor conductivity, and electrode degradation.^[8] Sulfur, being abundant and low-cost, contributes to reduced battery production costs and is environmentally benign relative to materials used in conventional LIBs. These characteristics make Li–S batteries attractive for high-energy applications in EVs and aerospace. The separator plays a critical role in Li–S battery performance and safety, acting both as a physical barrier between the electrodes and a control medium for polysulfide migration. It must exhibit excellent mechanical strength, porosity, flexibility, and ionic conductivity. Conventional polyolefin separators (e.g., polyethylene and polypropylene) offer low cost and chemical stability but perform poorly under elevated temperatures and in mitigating polysulfide diffusion.^[9] This can lead to severe capacity fading and increased safety risks.^[3] Advanced separators incorporating nonpolyolefin polymers, ceramics, nanofibers, or modified conventional membranes have been proposed to overcome these limitations.^[10] Materials such as nafion, polyacrylonitrile, poly(acrylic acid), polyvinylidene fluoride, polyethylene oxide (PEO), poly(methyl methacrylate), and polyetherimide have shown potential in improving mechanical and

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electrochemical properties. Polyethylene glycol (PEG)-based coatings enhance lithium-ion transport and electrolyte uptake by increasing hydrophilicity,^[11] while carbon-based additives like Super P, graphene, and carbon nanotubes improve electronic conductivity and suppress polysulfide migration.^[12] Reduced graphene oxide (rGO) enhances surface area and offers chemical adsorption for sulfur species.^[13] TiO₂ serves as an effective carbon-free coating for polysulfide immobilization and improved lithium kinetics,^[14] and boron nitride (BN) contributes to thermal management and lithium metal stabilization by blocking polysulfide diffusion.^[15] Electrolyte selection is equally crucial in Li-S battery design. Carbonate-based solvents, such as ethylene carbonate (EC), diethyl carbonate, propylene carbonate, and dimethyl carbonate, commonly used in LIBs, are generally incompatible with sulfur cathodes due to parasitic reactions with polysulfides. Nonetheless, they may be employed when sulfur is embedded in nanoporous structures or polymer matrices. Ether-based solvents such as dimethoxyethane (DME) and dioxolane are more suitable for Li-S systems owing to their chemical compatibility with polysulfides and ability to enable complete redox reactions.^[16] Among ethers, tetraethylene glycol dimethyl ether offers superior solvating power.^[17] Alternatives such as ionic liquids (ILs) and sulfones have also been explored, providing benefits such as thermal stability, low volatility, and suppression of polysulfide solubility. ILs like [TFSI]-based compounds improve battery cyclability and capacity retention by limiting polysulfide dissolution.^[18] Less volatile solvents such as NMP and dimethoxy sulfide (DMSO) are being evaluated for their favorable dielectric properties, low toxicity, and cost-effectiveness in Li-S configurations.

Thermal management in Li-S batteries is as critical as optimizing capacity and energy density. Heat generation correlates directly with energy loss and reduced material efficiency. To improve the electronic conductivity of sulfur cathodes, carbon additives are used; however, the sulfur-carbon composite poses safety concerns due to its flammability and reactivity at elevated temperatures.^[19] Excessive internal temperatures can lead to separator failure (e.g., melting of polyethylene or polypropylene) and trigger thermal runaway.^[20] Heat generation in Li-S batteries arises from three primary sources: 1) ohmic losses in the electrolyte, governed by the Nernst-Planck equation; 2) electrochemical reaction enthalpy at the electrode-electrolyte interface; and 3) overpotentials, including activation and concentration polarization. While polysulfide shuttling is a known challenge in Li-S systems, its effects were intentionally excluded in the present model to simplify the computational framework and isolate thermal behavior. This modeling decision enables focused analysis of heat generation and dissipation under controlled discharge conditions, thereby supporting the evaluation of material-dependent thermal responses. The study aims to investigate how thermal parameters influence overall cell performance, particularly with respect to power density, energy density, and energy storage.

2. Theoretical Approach

2.1. Model Development

The study has been performed over COMSOL Multiphysics Simulation software by designing a 1D hypothetical Li-S

model using the inbuilt electrochemistry -> tertiary current distribution, Nernst-Planck interface module, and heat transfer in fluids module in combination to design and simulate the thermal behavior of the cell, involving two types of studies namely, current distribution initialization and time-dependent study.

2.2. Mathematical Equations and Analysis

2.2.1. Total Current Density Due to Dissolved Ions

The equation for the general mass balance for the dissolved species in the electrolyte is given by the equation

$$\frac{\partial C_i}{\partial t} + \nabla \cdot N_i = R_{i,\text{tot}} \quad (1)$$

where N_i is the total flux of the i th diluted species ($\text{mol}/(\text{m}^2 \text{ s})$). Nernst-Planck equation determines the net flux in the electrolyte and accounts for the flux of charged solute species (ions) by diffusion, migration, and convection; these three modes collectively form the tertiary current distribution, and these individual modes form the first, second, and third term of the following equation

$$N_i = -D_i \nabla C_i - z_i \mu_{m,i} F C_i \nabla \phi_1 + c_i v = J_i + c_i v \quad (2)$$

where c_i represents the concentration of i th ion (mol m^{-3}), z_i represents the valency of i th ion, D_i is the diffusion coefficient ($\text{m}^2 \text{ s}^{-1}$), $\mu_{m,i}$ its mobility (s mol kg^{-1}), F denotes the Faraday constant (C mol^{-1}), ϕ_1 the electrolyte potential, v is the velocity vector (m s^{-1}), and J_i represents the molar flux relative to the convective transport.

$$J_i = -D_i \nabla C_i - z_i \mu_{m,i} F C_i \nabla \phi_1 \quad (3)$$

The total current density can be given by the summation of the flux of all species

$$i_1 = F \sum z_i N_i \quad (4)$$

Here, i_1 is the total current density in the electrolyte due to the movement of ions.

2.2.2. Modes of Heat Generation in the Cell

Heat sources are generated due to the following three irreversible voltage losses in an electrochemical cell: charge transport in the electrolyte, charge transport in the solid conductor materials, and activation overpotentials in the electrode reactions.

2.2.3. Joule Heating Due to Charge Transport

Joule heating source is created by transport across (solid) separator material and (fluid) electrolyte material, thus it is termed accordingly as

$$Q_{\text{JH}} = -(i_s \cdot \nabla \phi_s + i_l \cdot \nabla \phi_l) \quad (5)$$

For a separator and an electrolyte domain, this is the total electrochemical heat source. For porous electrode sources due to electrochemical reactions, it is added to the Joule heating. Heat transfer was modeled purely through conduction, as convection is negligible in the 1D geometry considered. The planar domain and boundary conditions assume zero velocity fields, which is appropriate for analyzing internal thermal dissipation under nonfluidic conditions. Hence, the COMSOL model accounts for Joule conductive heating only.

2.2.4. Heating Due to Electrochemical Reactions

The total heat balance equation for an electrochemical process can be given by the following equation

Heat generated = Total reaction enthalpy – Electrical energy leaving the system

This equation itself suggests that it accounts for the losses of the electrochemical cell

Using Faraday's laws for an electrode reaction, m , at the interface between the electron and ion-conducting phase, this corresponds to

$$Q_m = \left(\frac{\Delta H_m}{n_m F} - \left(\frac{\Delta G_m}{n_m F} - \eta_{m,\text{tot}} \right) \right) i_m \quad (6)$$

where ΔH_m is the enthalpy change of the reaction, and ΔG_m is the Gibbs free energy of the reaction, ΔG_m , defined as

$$\Delta G_m = \Delta H_m - T \Delta S_m \quad (7)$$

where ΔS_m is the net entropy change. The equilibrium potential is related to ΔG_m in the following equation

$$E_{\text{eq},m} = - \frac{\Delta G_m}{n_m F} \quad (8)$$

$\eta_{m,\text{tot}}$ is the total overpotential, which is defined as

$$\eta_{m,\text{tot}} = \phi_s - \phi_l - E_{\text{eq},m} \quad (9)$$

By the relation

$$\frac{\partial E_{\text{eq},m}}{\partial T} = \frac{\Delta S_m}{n_m F} \quad (10)$$

the local heat source due to the electrochemical conversion process becomes

$$Q_m = \left(\phi_s - \phi_l - E_{\text{eq},m} + T \frac{\partial E_{\text{eq},m}}{\partial T} \right) i_m \quad (11)$$

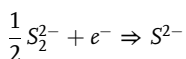
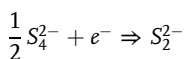
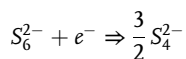
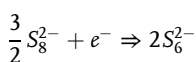
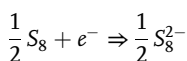
The total heat source due to the electrochemical reactions, Q_{EC} , for an electrode surface is the sum of all individual heat sources of the electrode reactions according to

$$Q_{\text{EC}} = \sum_{m=0}^n Q_m \quad (12)$$

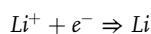
3. Results and Discussions

3.1. 1-D Li-S Electrochemical Cell Model

The design of the model involves 1D geometry, as shown in **Figure 1**, developed using the electrochemical heating multiphysics interface in COMSOL Multiphysics Version 6.2. The parameters used in the study have been listed in **Table 1**. The separator and the positive (cathode during discharge) porous electrode are defined as domains in the geometry. The leftmost boundary represents the negative lithium metal electrode, and the rightmost boundary represents the positive metal current collector. The electrolyte mass and charge transport are modeled using the Nernst–Planck equations with electroneutrality. The charge transfer reactions are defined using the Nernst equation in combination with the Butler–Volmer equation, assuming the law of mass action. In addition to the earlier six polysulfide species, the species Li^+ and A^- are included in the electrolyte, where A^- is the counter anion of the lithium salt. The same species are also present in the separator domain. Two precipitation–dissolution (non-Faradaic) reactions involving the solid species S_8 and Li_2S are included in the model (in both domains). The model is based on the Nernst–Planck model, which assumes electroneutrality for which it is assumed that one of the ions is independent of its concentration, which is reasonably the counter anion $[\text{A}^-]$ because it has a relatively high concentration and behaves as a spectator ion, while the electrode reactions take place. The tertiary current distribution, Nernst–Planck interface, has been used wherein the model includes the separator and the porous electrode, and it has been defined that the reaction takes place in their domain. In the separator as well as the electrolyte, it has been only the precipitation equation that leads to the precipitation of Li_2S and S_8 ; however, it is a point to note that five sequential reactions have occurred correspondingly in the cathode. The following five charge transfer reactions and six participating polysulfide species are considered in the positive electrode domain



The electrode surface reaction has been defined for the Li anode in the first boundary as follows



The model analysis and evaluations generate certain parameters within “Tertiary Current Distribution” interface, which is utilized by “Electrochemical Heating Multiphysics” interface to calculate the Q_{JH} , which is evaluated as the summation of

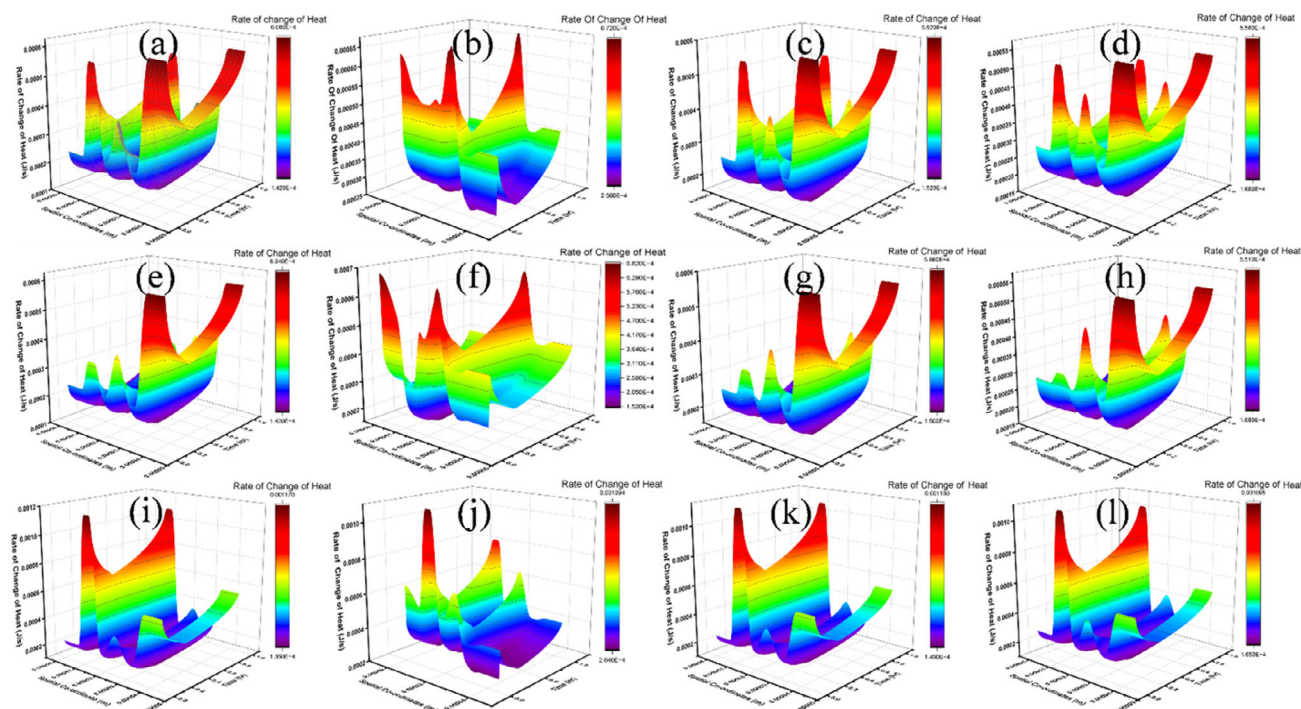


Figure 1. 3D contoured surface representing variation in rate of change of heat across the cell length at constant discharge rate of a) BN–acetonitrile, b) BN–DME, c) BN–DMOE, d) BN–DMSO, e) TiO₂–acetonitrile, f) TiO₂–DME, g) TiO₂–DMOE, h) TiO₂–DMSO, i) PEG–acetonitrile, j) PEG–DME, k) PEG–DMOE, and l) PEG–DMSO.

three individual heat sources namely, Q_{ohmic} , $Q_{reaction}$, and $Q_{overpotential}$

$$Q_{JH} = Q_{ohmic} + Q_{reaction} + Q_{overpotential} \quad (13)$$

To evaluate the above three quantities, COMSOL will require temperature as a function of time, which the multiphysics interface will be getting through the “Heat Transfer in Fluids,” which requires a model as listed in Table 1. An analytical function has been defined over the model domain to specify various parameters such as density ($\rho(x)$), specific heat capacity at constant pressure ($C_p(x)$), and thermal conductivity ($k(x)$). These parameters have been assigned across the domain to represent multiple materials—including the separator, electrolyte, and cathode—and are detailed in the Experimental Section. **Table 2** lists all the material parameters that have been required throughout the model.

The model has been designed with certain limitations that define the boundary of the model, which include: 1) the geometry of the model, being 1D is defined over a length, i.e., from 0 (origin) to 45 μm , thus, all parameters are bound within this frame. 2) The operating voltage of the model is 1.7 V, if during model simulation, the voltage dips below 1.7 V, the processing terminates. 3) The current density drawn from the cell for a fixed discharge rate is given constant which is at 1C discharge rate, and the current density is 6.13 A m^{-2} . 4) To focus on the primary thermal behavior of different separator–electrolyte combinations, the model omits cross-electrode polysulfide transport (shuttling effect). While this exclusion limits direct applicability to long-cycle

degradation behavior, it enables isolated thermal benchmarking and reveals intrinsic material-dependent thermal profiles.

3.2. Simulation Result and Analysis

The model has been optimized by computing the performance parameters and comparing the results of the simulation for varying configurations. Each configuration is designed by taking a particular combination of separator and electrolyte material. The possible pairs include BN–DME, BN–dimethoxy ethane, BN–DMSO, BN–acetonitrile, TiO₂–DME, TiO₂–dimethoxy ethane, TiO₂–DMSO, TiO₂–acetonitrile, PEO–DME, PEO–dimethoxy ethane, PEO–DMSO, and PEO–acetonitrile. Thus, the result obtained is aimed at realizing an optimized configuration that manages to generate a decent electrical output without much abnormal thermal behavior, thereby maintaining a suitable environment-friendly and safety-ensured design. The first plotted study inspects the rate of change of heat as shown in **Figure 2** and not exactly the amount of heat because the model is designed to be highly constrained in terms of its size which makes it a very indeterministic property. Also, it has been aimed to design a model that has limitations to the rise of temperature rather than the heat stored because thermal runaway in battery occurs at a point of temperature where self-sustaining reactions get triggered due to which the temperature further keeps on increasing leading the battery to vent, catch fire, or explode due to which the temperature marks the second point of study quite reasonably given in **Figure 3**. Also, the internal resistance is taken into consideration as previously discussed on how it affects

Table 1. The parameters defined in COMSOL multiphysics for the modelling of 1D Li-S battery.

Sr.No	Name	Expression	Description
1)	T	30 [degC]	"Temperature"
2)	Av_0	132 762 [m ² /m ³]	"Specific surface area"
3)	Ksp_Li2S_s	1e2 [mol ³ *m ⁻⁹]	"Solubility product"
4)	Ksp_S8_s	19 [mol/m ³]	"Solubility product"
5)	L_sep	25e-6 [m]	"Separator thickness"
6)	c_A_1m_ref	1000 [mol/m ³]	"Reference concentration, salt anion"
7)	c_S_2m_ref	8.27e-10 [mol/m ³]	"Reference concentration, S ⁻²ⁿ "
8)	c_S8_2m_ref	0.18 [mol/m ³]	"Reference concentration, S ₈ ⁻²ⁿ "
9)	c_S8_ref	19 [mol/m ³]	"Reference concentration, S ₈ "
10)	c_S6_2m_ref	0.32 [mol/m ³]	"Reference concentration, S ₆ ⁻²ⁿ "
11)	c_S4_2m_ref	0.02 [mol/m ³]	"Reference concentration, S ₄ ⁻²ⁿ "
12)	c_S2_2m_ref	5.23e-7 [mol/m ³]	"Reference concentration, S ₂ ⁻²ⁿ "
13)	c_Li_1p_ref	1001 [mol/m ³]	"Reference concentration, Li ⁺ "
14)	k_S8_s	5 [1/s]	"Rate constant, S ₈ (s) deposition-dissolution"
15)	k_Li2S_s	3.45e-5 [m ⁶ *mol ⁻² /s]	"Rate constant, Li ₂ S (s) deposition-dissolution"
16)	Vm_S8_s	1.24e-4 [m ³ /mol]	"Molar volume, S ₈ (s)"
17)	Vm_Li2S_s	2.4e-5 [m ³ /mol]	"Molar volume, Li ₂ S (s)"
18)	M_S8_s	8×32.07 [g/mol]	"Molar mass, S ₈ (s)"
19)	M_Li2S_s	(32.07+2×6.941) [g/mol]	"Molar mass, Li ₂ S (s)"
20)	eps_S8_s_sep_0	1e-12	"Initial volume fraction in separator of S ₈ (s)"
21)	eps_Li2S_s_sep_0	1e-7	"Initial volume fraction in separator of Li ₂ S (s)"
22)	eps_S8_s_pos_0	0.166	"Initial volume fraction in electrode of S ₈ (s)"
23)	eps_Li2S_s_pos_0	1e-7	"Initial volume fraction in electrode of Li ₂ S (s)"
24)	i0_5_ref	2e-9 [A/m ²] *0+2e-7 [A/m ²]	"Exchange current density at reference concentrations, reaction 5"
25)	i0_4_ref	2e-4 [A/m ²]	"Exchange current density at reference concentrations, reaction 4"
26)	i0_3_ref	0.02 [A/m ²]	"Exchange current density at reference concentrations, reaction 3"
27)	i0_2_ref	0.02 [A/m ²]	"Exchange current density at reference concentrations, reaction 2"
28)	i0_1_ref	1.9 [A/m ²]	"Exchange current density at reference concentrations, reaction 1"
29)	i0_Li_ref	0.5 [A/m ²]	"Exchange current density at reference concentrations, Li/Li ⁺ reaction"
30)	Eeq_5_ref	2.01[V]-R_const*T/F_const*log((c_S2_2m_ref/1[M]) ^{-1/2})	"Equilibrium potential at reference concentrations, reaction 5"
31)	Eeq_4_ref	2.03[V]-R_const*T/F_const*log((c_S4_2m_ref/1[M]) ^{-1/2})	"Equilibrium potential at reference concentrations, reaction 4"
32)	Eeq_3_ref	2.23[V]-R_const*T/F_const*log((c_S6_2m_ref/1[M]) ⁻¹)	"Equilibrium potential at reference concentrations, reaction 3"
33)	Eeq_2_ref	2.35[V]-R_const*T/F_const*log((c_S8_2m_ref/1[M]) ^{-3/2})	"Equilibrium potential at reference concentrations, reaction 2"
34)	Eeq_1_ref	2.41[V]-R_const*T/F_const*log((c_S8_ref/1[M]) ^{-1/2})	"Equilibrium potential at reference concentrations, reaction 1"
35)	Eeq_Li_ref	0[V]-R_const*T/F_const*log((c_Li_1p_ref/1[M]) ⁻¹)	"Equilibrium potential at reference concentrations, Li/Li ⁺ reaction"
36)	epsl_sep_0	0.5	"Electrolyte volume fraction of separator without deposits"
37)	epsl_pos_0	0.7	"Electrolyte volume fraction of positive electrode without deposits"
38)	L_pos	20e-6 [m]	"Electrode thickness"
39)	sigma_s	1 [S/m]	"Electric conductivity, electrode"
40)	D_A_1m	3.5e-12 [m ² /s]	"Diffusion coefficient, salt anion"
41)	D_S_2m	0.88e-12 [m ² /s]	"Diffusion coefficient, S ⁻²ⁿ "
42)	D_S8_2m	3.5e-12 [m ² /s]	"Diffusion coefficient, S ₈ ⁻²ⁿ "
43)	D_S8	0.88e-11 [m ² /s]	"Diffusion coefficient, S ₈ "
44)	D_S6_2m	3.5e-12 [m ² /s]	"Diffusion coefficient, S ₆ ⁻²ⁿ "
45)	D_S4_2m	1.75e-12 [m ² /s]	"Diffusion coefficient, S ₄ ⁻²ⁿ "
46)	D_S2_2m	0.88e-12 [m ² /s]	"Diffusion coefficient, S ₂ ⁻²ⁿ "
47)	D_Li_1p	0.88e-12 [m ² /s]	"Diffusion coefficient, Li ⁺ "

Table 1. Continued.

Sr.No	Name	Expression	Description
48)	rho_S8_s	$M_{S8_s}/V_{m_S8_s}$	"Density, S ₈ (s)"
49)	rho_Li2S_s	$M_{Li2S_s}/V_{m_Li2S_s}$	"Density, Li ₂ S (s)"
50)	C	1 C-rate	"C-rate"
51)	Cap	$16 \cdot (c_{S8_ref} + \epsilon_{S8_s_pos_0} / V_{m_S8_s}) \cdot L_{pos} \cdot F_{const}$	"Cell capacity (theoretical)"
52)	Cap_nom	$3.4 [A \cdot h] / A_{cell}$	"Cell capacity (nominal)"
53)	A_cell	0.28 [m ²]	"Cell area"
54)	i_1C	$Cap_nom / 1 [h]$	"1C current density"

Table 2. The essential material properties during the model study and assessment.

Material	Thermal Conductivity	Heat Capacity at Constant Pressure	Density
Boron nitride, BN	30 W/(m.K)	1610 J/(kg.K)	1900 kg m ⁻³
Rutile, TiO ₂	11.8 W/(m.K)	697 J/(kg.K)	4050 kg m ⁻³
Polyethylene glycol, PEG	0.3 W/(m.K)	2800 J/(kg.K)	1150 kg m ⁻³
Dimethyl ether, DME	0.015616 W/(m.K)	1443.1 J/(kg.K)	1970 kg m ⁻³
Dimethoxy ethane, DMOE	0.161 W/(m.K)	2134 J/(kg.K)	868.3 kg m ⁻³
Dimethyl sulfoxide, DMSO	0.1567 W/(m.K)	2008.32 J/(kg.K)	1023 kg m ⁻³
Acetonitrile	0.21 W/(m.K)	2223.76 J/(kg.K)	798.12 kg m ⁻³

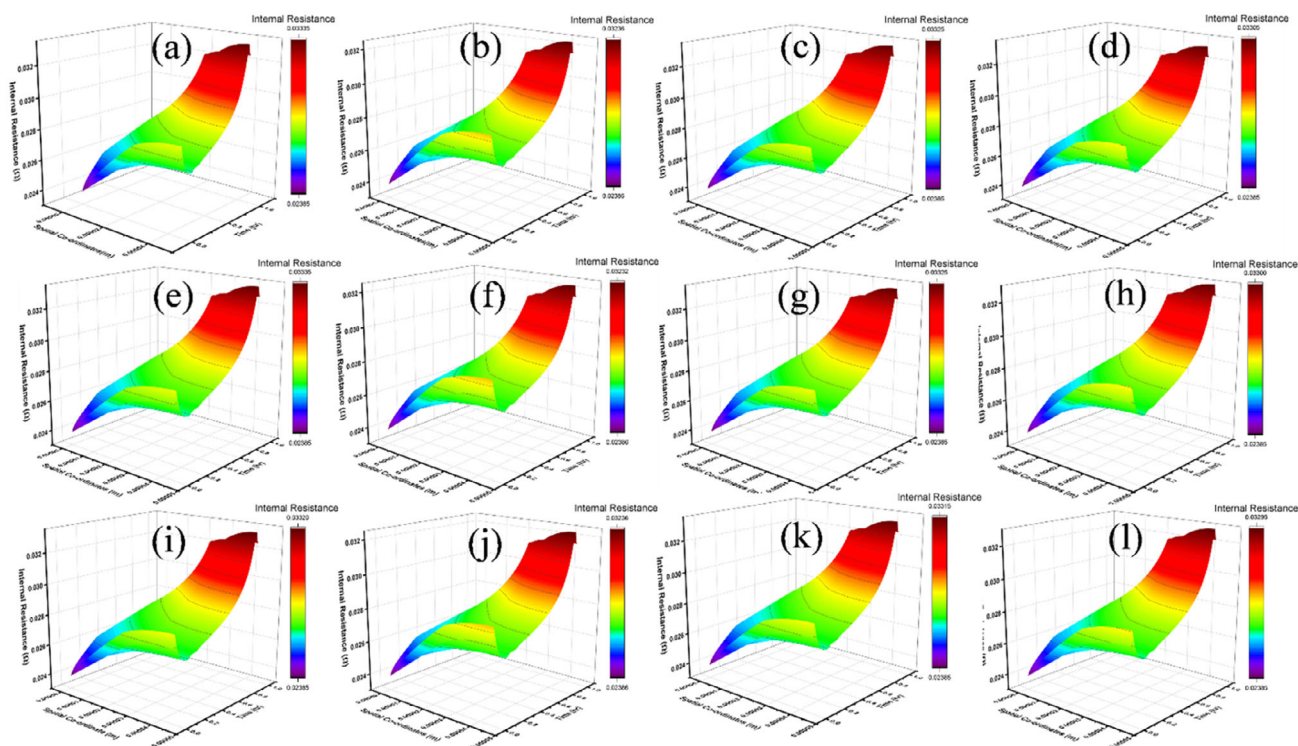


Figure 2. Schematic representation of internal resistance in a surface showing the variation of position along the length of the cell and of time duration at a constant discharge rate of a) BN-acetonitrile, b) BN-DME, c) BN-DMOE, d) BN-DMSO, e) TiO₂-acetonitrile, f) TiO₂-DME, g) TiO₂-DMOE, h) TiO₂-DMSO, i) PEG-acetonitrile, j) PEG-DME, k) PEG-DMOE, and l) PEG-DMSO.

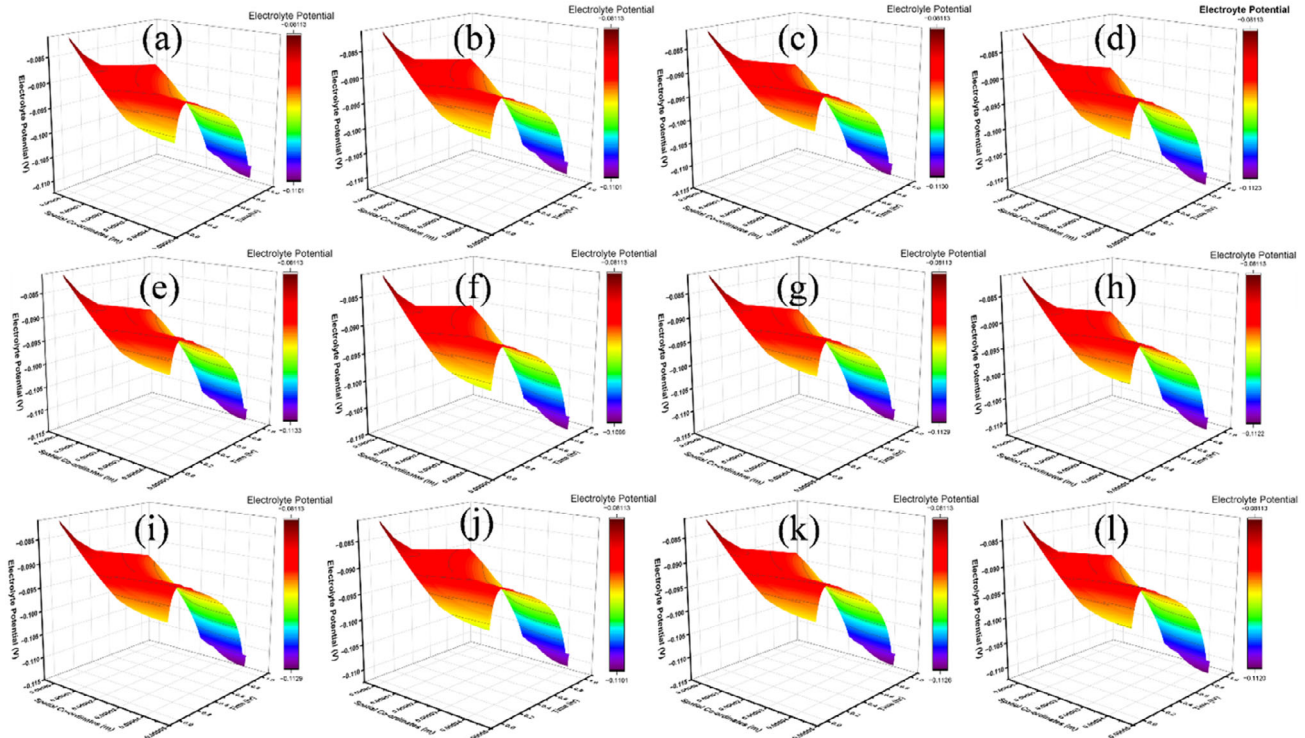


Figure 3. 3D Schematic surface representation for electrolyte potential for all the combinations of separator–electrolyte in model a) BN–acetonitrile. b) BN–DME. c) BN–DMOE. d) BN–DMSO. e) TiO₂–acetonitrile. f) TiO₂–DME. g) TiO₂–DMOE. h) TiO₂–DMSO. i) PEG–acetonitrile. j) PEG–DME. k) PEG–DMOE. l) PEG–DMSO.

the discharge rate of the electrochemical cell. Thus, the internal resistance variation across the length of the cell provides a better insight into the cumulative performance study. Keeping in mind the thermal and electrical output, it is displayed in **Figure 4**. Later light has been thrown over the obtained result of electrolyte potential using different combinations. Using these inspection parameters, the best configuration possible has been declared.

3.3. Effect on Rate of Change of Heat

The heat of the electrochemical 1D cell in the electrolyte and separator medium has been calculated using the relation

$$dQ = m \cdot C_p \cdot dT$$

$$\frac{dQ}{dx} = m \cdot C_p \cdot \frac{dT}{dx} \Rightarrow \frac{dQ}{dt} = V_{\text{elemental}} \cdot \rho(x) \cdot C_p(x) \cdot \frac{dT}{dt}$$

It is known that the elemental volume can be given by

$$V_{\text{elemental}} = A_{\text{cell}} \cdot x_{\text{elemental}}$$

$$x_{\text{elemental}} = \Delta x \approx 1e-7 (\text{assumed})$$

Parameters like density and heat capacity at constant pressure have been defined over the domains in the format of an analytical function. As can be seen from the resultant simulation in Figure 2, all configurations using PEG as a separator material

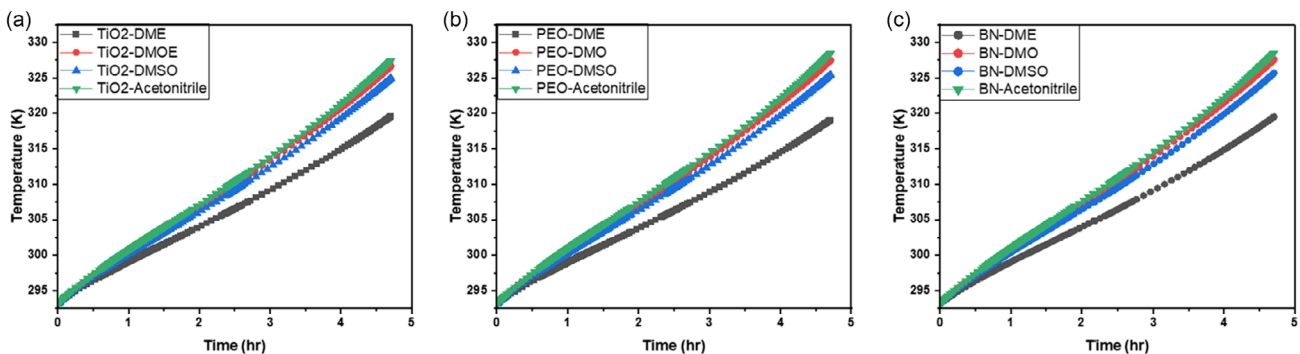


Figure 4. Temperature variation in electrolytic medium a) for separator: BN, b) for separator: PEG, and c) for separator: TiO₂.

have fairly high values in comparison to the one using BN and TiO_2 . Also, in cases of BN–DME, TiO_2 –DME, BN–acetonitrile, and TiO_2 –acetonitrile are, respectively, high in comparison to other combinations of BN and TiO_2 , which seem less sensitively variable. Further inspection was carried out on the remaining combinations: BN–dimethoxyethane (DMOE), BN–DMSO, TiO_2 –DMOE, and TiO_2 –DMSO. The rate of change of heat is found to be minimal in the DMSO configuration of BN and TiO_2 . The DMOE configuration does not exhibit very high thermal variation and can be considered as well; however, in terms of the rate of heat change, the TiO_2 –DMSO configuration shows the lowest value. It is also notable that in the case of DMOE configurations, there is a clear irregularity in the 3D surface formed, while DMSO configs have lesser peaks, indicating a lesser no. of points of localized heating in the cell.

Another possible factor of consideration could be internal resistance.

3.4. Internal Resistance Variation

It is already known that the internal resistance affects the discharge rate. A high value is expected for discharge rate as there can be times when the vehicle is needed to sustain a heavy load, maintaining the speed of the vehicle to be constant.

In Figure 3, the variation along the cell length and different time instants during the period of cell operation is mostly similar. Only the difference lies in the magnitude. The reason for this can be understood from the fact that internal resistance is simply given by

$$R_{\text{int}} = \frac{V_{\text{electrolyte}}}{(i_{0.2C}) \cdot A_{\text{cell}}} \quad (14)$$

In this formula, it can be seen how R_{int} is directly proportional to the electrolyte potential that has been plotted for each material

configuration as shown in Figure 4. For every other separator material, the value of internal resistance does not change as such because it is mainly dependent on electrolytes to conduct electricity across electrodes. The model has neglected the separator engineering to establish the complete circuit through it and hence taken as a uniform for every separator material.

Thus, among acetonitrile, DME, DMOE, and DMSO, it can be seen that DME always offers the least resistant path, which was obvious, keeping in mind that it had the best electrical conductivity among its competitors. Even among BN–DME, TiO_2 –DME, and PEG–DME, the least resistant path is provided by PEG–DME as obtained in the result.

The next parameters that are to be inspected before declaring any particular configuration to be the best, based on the study and purpose, are temperature and cell voltages.

3.5. Temperature Attained by Cell

The temperature of the model is calculated using the “Multiphysics” library and has been generated as one of the state variables of the model; thus, it does not need to be evaluated manually in terms of a mathematical expression. The model is isothermal; therefore, all points on the cell domain attain the same temperature. An isothermal approximation was adopted considering the short domain length ($\approx 45 \mu\text{m}$), low heat generation rates, and the dominance of thermal conduction. This assumption is appropriate for evaluating the intrinsic thermal behavior of materials under steady-state, controlled-discharge conditions.

It can be easily interpreted from the obtained result, as shown in Figure 5, that there are no significant noticeable changes in curves as the model opts for different separator materials; however, it can be seen that the trend in changes of temperature curve for different electrolytic medium is homogeneous for

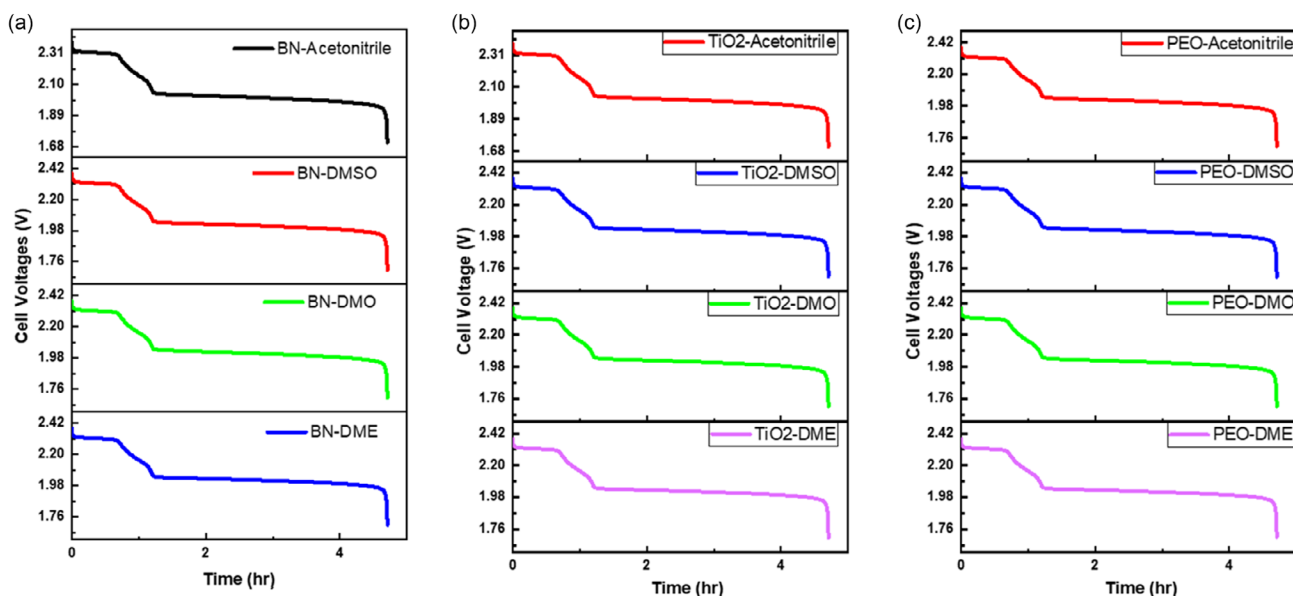


Figure 5. Output cell voltage due to Li–S battery with different compositions a) with BN as a separator, b) with PEG as a separator, and c) with TiO_2 as a separator.

varying separator material which supports the fact that there is not much change in temperature due to change in separator material because the material only forms the layer over the separator and not the actual separator mass. On keen observation though, this change can be marked that in BN and TiO_2 , the plots are more spread apart than in PEG. Also, max temperature is achieved in BN than PEG and TiO_2 ; however, this variation is very fine. However, there is a remarkable difference in the case of electrolytes. For DME, the rise in temperature is found to be the least and highest in the case of acetonitrile, and this trend is common for every separator material. Thus, DME seems to be the first choice, then DMSO, DMOE, and lastly acetonitrile

3.6. Cell Voltages as Electrical Output of Cell

It is evident that the cell voltage output, as shown in **Figure 6**, is purely dependent on the cell reaction, and electrochemical equations are involved to theoretically determine the cell voltage. The model has negated the effect of polysulfide deposition (shuttling effect) as discussed earlier; therefore, the engineered cell, irrespective of the material, assumes a complete cutoff of polysulfide deposition. Thus, as a result, all combination achieves similar cell voltage.

This assumption predominates the purpose to solely look out for a better cell which ensures higher thermal safety along with satisfactory electrical output.

Now, enough factors are there to declare the most suitable configuration, and it will be expected to achieve the better thermal performance of the system due to which temperature rise and rate of change of heat are the prime concerns, while internal resistance is the top second priority. Thus, based on the result obtained, it has been considered that BN–DME is the most suitable model, as it has the minimum. However, this conflicts with the rate of change assumption, but the minimum temperature offered and significantly less resistant path offered without much increment in rate of change of heat, which promotes it to be the most prominent among the rest. But let's take the environmental

factors for once, as there is yet another aim of the model to prepare eco-friendly design.

Separators in Li–S batteries play a critical role in preventing short circuits while allowing lithium-ion flow. Their sustainability can be assessed based on production impact, recyclability, and stability.

3.7. Polyethylene Glycol

PEG is derived from ethylene glycol, a petroleum-based chemical. However, it is biodegradable and used in pharmaceuticals, showing promise as an eco-friendly material. Studies show that PEG-based separators can enhance battery life while maintaining environmental compatibility. PEG can be synthesized in a green manner, and it has relatively low toxicity. Its polymeric nature can also be adapted to recycling processes.^[21]

3.8. Boron Nitride

BN is a highly stable ceramic material. While not biodegradable, it offers excellent thermal and chemical stability, enhancing the longevity of the battery and reducing the need for frequent replacement. BN requires significant energy to produce but offers minimal toxicity and long-term environmental damage. It is recyclable but requires specialized processes, reducing its overall environmental friendliness.^[22]

3.9. Titanium Dioxide (TiO_2)

Extraction and refinement of TiO_2 have significant environmental costs. TiO_2 has moderate toxicity concerns related to the production process, though it is considered a relatively safe material. It is not easily biodegradable, and recycling TiO_2 from battery separators remains complex.

Electrolytes serve as the medium for ion transfer and impact battery safety, performance, and environmental footprint. Their

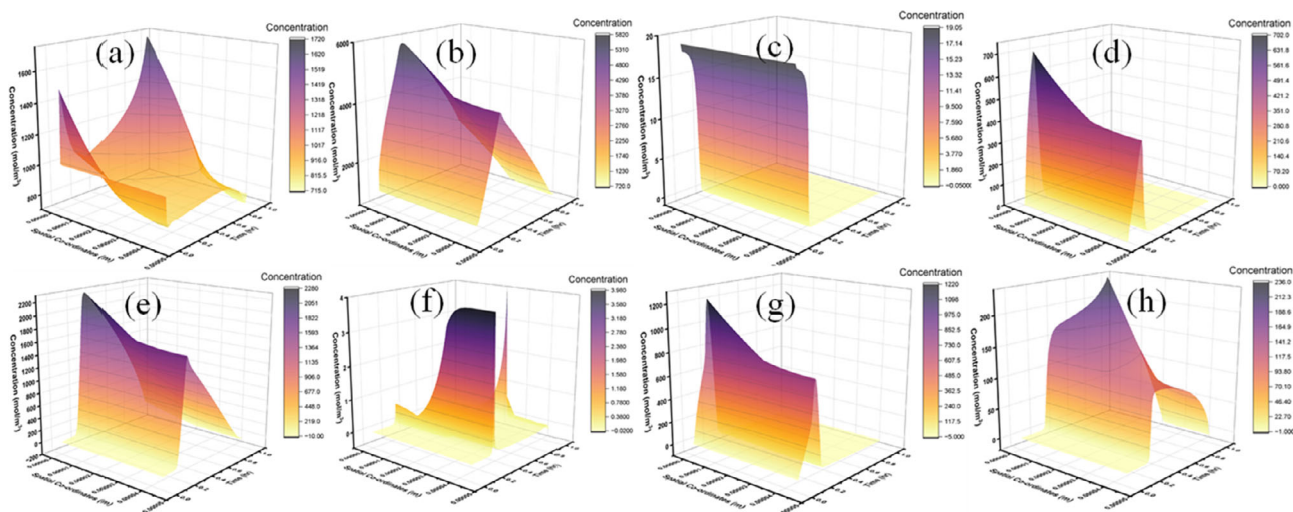


Figure 6. Concentration profile of all the ions involved in the electrochemical chain reactions, i.e., a) for A^- , b) for Li^+ , c) for S_8 , d) for S_8^{2-} , e) for S_4^{2-} , f) for S^{2-} , g) for S_6^{2-} , and h) for S_2^{2-} .

sustainability depends on the toxicity, volatility, and renewability of the chemicals.

3.10. Dimethyl Ether

DME is considered a clean fuel and can be produced from renewable resources, including biomass. It is nontoxic and eco-friendly. It exhibits high ionic conductivity in Li-S batteries and reduces polysulfide shuttle effects, extending battery life. Low greenhouse gas emissions and nontoxic degradation products make DME the most environmentally friendly choice. It is also biodegradable.^[23]

3.11. Dimethoxy Ethane

DME is a solvent with moderate environmental impact. It performs well in batteries but has slightly higher toxicity and flammability risks than DME. While not biodegradable, DME is relatively safe compared to many solvents. However, its synthesis from petroleum can contribute to carbon emissions.^[24]

3.12. DMSO

DMSO is a sulfur-containing compound, meaning its production can release harmful sulfur oxides (SO_x). It has been found to provide high performance in Li-S batteries, but its environmental sustainability is limited. DMSO is less eco-friendly due to its sulfur content and the potential formation of hazardous byproducts.^[25]

3.13. Acetonitrile

Acetonitrile is a solvent that can be used in many electrochemical systems, including Li-S batteries. However, it is highly toxic and poses environmental hazards during its production and disposal. Acetonitrile is considered harmful to both health and the environment. While effective in terms of battery performance, its toxicity makes it the least environmentally friendly option.^[26,27]

Based on the sustainability and environmental impact of the separators and electrolytes, the best configurations for the Li-S battery would involve PEG-DME, followed by PEG-DMOE. The least favorable would be combinations involving acetonitrile, especially with TiO₂ as a separator. However, it is known that the thermal and electrochemical performance of PEG is not up to the mark, and hence, there is a need to choose a trade-off to accept the BN-DME model to be the best choice, as per the analysis.

The Li-S battery using boron nitride to design its separator and dimethyl ether to be used as an electrolyte in the model needs further analysis based on various other relevant parameters. The model involves studying various battery performance-specific parameters to overview the overall performance of this configured system. The concentration profile of all the ions that have participated in the reaction is depicted in Figure 6.

The next parameters studied are the energy stored and power density of the given configuration, which essentially account for better battery performance. For EV, the model requires high power density because it is required so as to withdraw large amounts of energy sometimes for heavy applications; however, specific power density is important to constrain that same prowess in a small size by mass or volume. A plot for power density

and energy stored, as shown in Figure 7 and Figure 8, has been developed with the software.

Another significant parameter that has been analyzed is the cell voltage vs capacity plot. It is a fundamental diagnostic tool for evaluating the battery's current state, understanding its aging process and optimizing charging-discharging protocols for improved performance and longevity. The graph shown in Figure 9 depicts the obtained cell voltage vs capacity curve for the given plot.

The Ragone plot, obtained from the optimized Li-S cell model developed in this study with a focus on reduced thermal instability, was thoroughly analyzed. Figure 10 illustrates a comparative performance assessment between the simulated results generated using the present COMSOL model and experimentally reported data by Vivanco and Rodríguez-Monroy.^[28] The referenced study discusses graphene-enhanced Li-S batteries and highlights the advantages offered by graphene's intrinsic

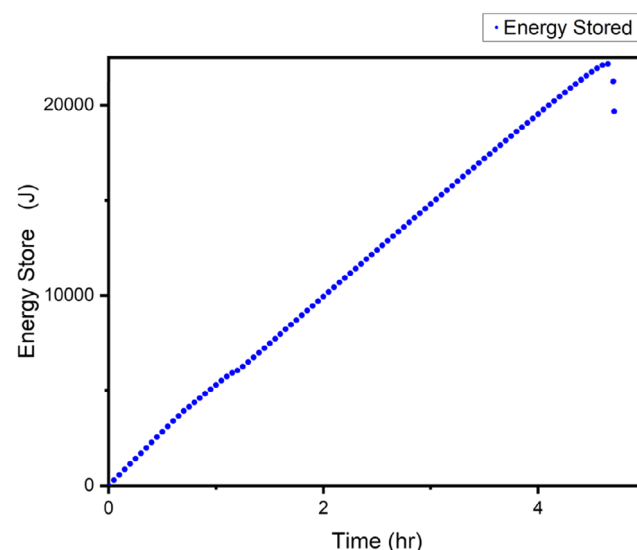


Figure 7. Energy stored in Li-S battery with BN as a separator and dimethyl ether as electrolyte.

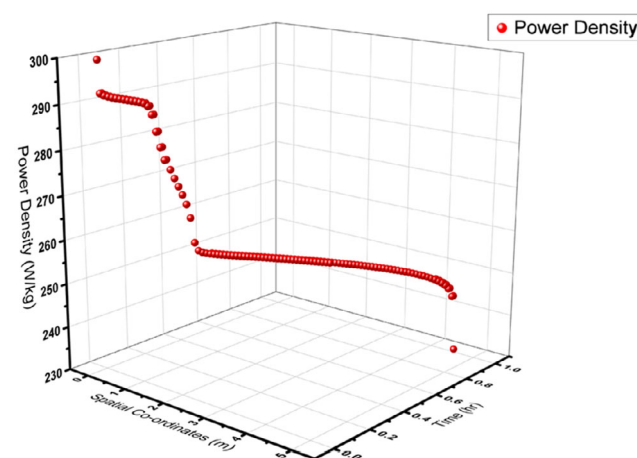


Figure 8. Power density for BN-DME configuration in Li-S battery.

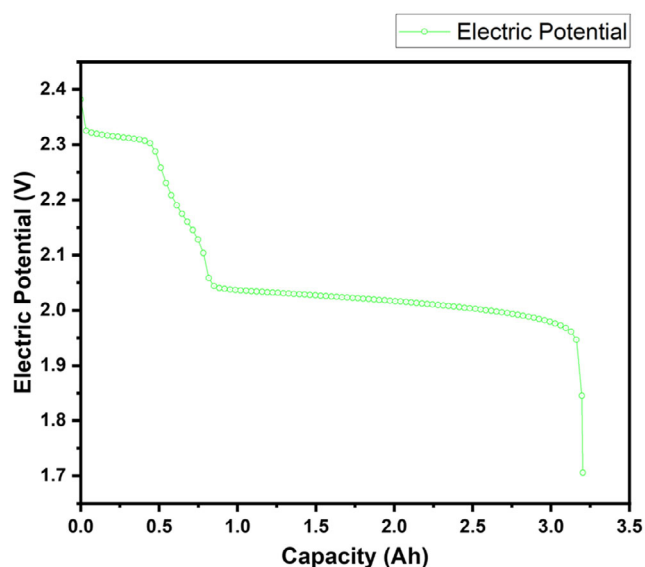


Figure 9. Cell voltage versus capacity curve for optimized Li-S cell configuration.

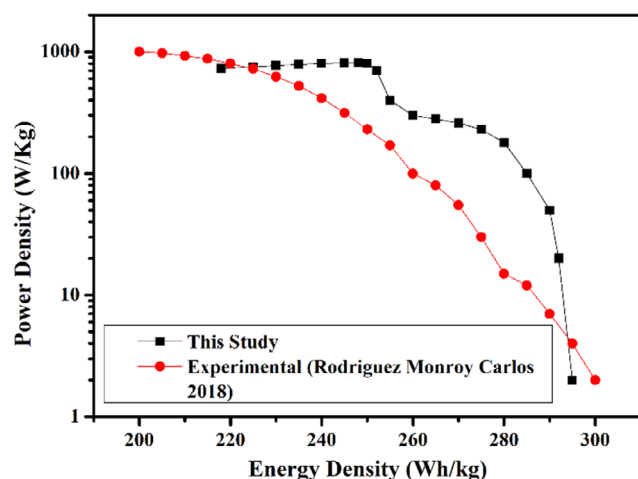


Figure 10. Ragone plot for a Li-S battery with BN-DME configuration, showing the trade-off between energy density and power density as obtained from the COMSOL model. The BN-DME pair demonstrates high thermal stability and electrochemical efficiency, with peak power density above 800 W kg^{-1} and energy density up to $\approx 295 \text{ Wh kg}^{-1}$.

properties—such as high surface area, superior charge carrier mobility, and notable chemical stability. According to the findings, hybrid Li-S systems incorporating graphene and carbon nanotubes are capable of achieving energy densities up to 400 Wh/kg and power densities nearing $10\,000 \text{ W/kg}$.

In contrast, the simulated configuration developed in this work, which excludes graphene integration, attains a peak energy density of $\approx 290 \text{ Wh kg}^{-1}$ and a power density close to 810 W kg^{-1} . These results are in line with typical values observed in conventional Li-S battery systems. The model effectively captures the characteristic trade-off between energy and power density, closely aligning with the experimental trend in the mid-energy range ($\approx 220\text{--}290 \text{ Wh kg}^{-1}$). Slight deviations

are attributed to idealized modeling conditions, including the omission of polysulfide shuttling effects and the assumption of uniform temperature distribution.

4. Conclusion

This study investigated the thermal and electrochemical performance of twelve separator–electrolyte combinations in Li–S batteries using a 1D COMSOL Multiphysics model. Among these, the BN–DME configuration emerged as the most effective, exhibiting a low peak temperature of 319.52 K , high thermal stability, and minimal internal resistance (0.0326Ω). It also achieved a maximum voltage of 2.38 V and a stable plateau at 2.025 V —among the highest recorded. While the TiO_2 –DMSO pair showed the slowest temperature rise rate, it underperformed in voltage and resistance metrics. BN–DME also offers a favorable environmental profile compared to acetonitrile- or TiO_2 -based systems, supporting its sustainability. Though mechanical durability and long-term cycling were not simulated, prior studies indicate BN–DME provides high ionic conductivity, better heat dissipation, and partial suppression of polysulfide shuttling—factors beneficial for cycle life and structural reliability. The model also delivered energy storage and power density results, with energy peaking at 22.157 kJ and stepwise power behavior indicating consistent electrochemical activity. A Ragone plot comparison with experimental data confirmed the model's predictive reliability. Overall, the BN–DME configuration offers an optimal balance of safety, thermal performance, and efficiency for advanced Li–S battery applications.

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Conflict of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Keywords

energy densities, joule heating, self-discharge, shuttling effects, specific capacities, thermal runaways

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